

BlockSign Verify

Advanced Video Liveness Detection. Privacy-first, real-time identity verification combining document scanning, multi-factor liveness detection, AI-driven questionnaires, and speech verification into a single seamless workflow. **No video, images, or biometric data stored.**

Privacy-First

Zero Storage

Real-Time

AI-Powered

GDPR Ready

CCPA Compliant

The AI Shift

Artificial intelligence is advancing at a parabolic rate. Deepfakes, synthetic voices, and AI-generated identities are becoming indistinguishable from real humans. The question is no longer *"what can AI do?"* but *"how do we prove a human is real?"*

Every critical digital process will require verified human presence. Not just passwords. Not just biometrics. A real-time, privacy-preserving proof that a living human is behind the screen.

Where Proof of Human Is Critical

Login & Authentication

Verify a real person is signing in, not an AI bot or deepfake replay attack.

eSign Documents

Ensure the individual digitally signing contracts and legal documents is a verified human.

Release of Transactions & Funds

Authorize high-value financial transactions with confirmed human presence before funds move.

Apply for Funds or Credit

Confirm human identity when applying for loans, grants, or financial products to prevent synthetic fraud.

Access Restricted Systems

Gate access to sensitive systems, classified data, or privileged operations behind live human verification.

Regulatory Compliance

Meet KYC, AML, and identity verification mandates with auditable proof of human presence.

Product Overview

BlockSign Verify combines four core capabilities into a unified verification platform that confirms human identity in under two minutes.

Prechecks & Quality Gates

- Video quality analysis and resolution validation
- Face detection and face-in-frame positioning
- Lighting assessment for optimal capture
- Video stream and FPS monitoring

Data Extraction & OCR

- ID cards, passports, driver licenses
- OCR extraction via Tesseract / AI
- Auto-detection of document type
- Front and back capture with flip delay

Chatbot Verification

- Interactive questionnaire for onboarding
- Data aggregation and cross-referencing
- Document data validation via live responses
- Customizable question catalogs

Liveness Detection

- Multi-factor video and voice analysis
- Real-time confidence scoring
- Anti-spoofing with 3D parallax
- 8-factor weighted liveness score

Voice & Speech Liveness Detection

Users are presented with **5 random words** that must be spoken **in the given order**. The system verifies speech through multiple channels simultaneously.

How It Works

- 5 random words displayed in sequence
- Words must be spoken in the exact given order
- Real-time speech recognition via Web Speech API
- Audio volume monitoring with visual feedback
- Cross-validation with lip movement on video feed

Why It Matters

- Defeats pre-recorded audio attacks
 - Random word order prevents memorized scripts
 - Volume analysis detects synthetic audio playback
 - Lip-sync verification adds a visual layer
 - Combined with video liveness for multi-modal proof
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Video & Face Liveness Detection

Multi-signal analysis from the video feed detects genuine human presence and defeats spoofing attempts including printed photos, screen replays, and 3D masks.

Eye Blink Rate Detection

Natural blink patterns measured via Eye Aspect Ratio (EAR) analysis from 68-point facial landmarks. Detects absence of blinking in static images and unnatural blink rates.

Heartbeat Detection (rPPG)

Remote photoplethysmography extracts pulse signal from subtle skin color changes in the forehead region. FFT analysis isolates cardiac rhythm in the 48-180 BPM range.

3D Parallax Analysis

Detects natural depth variation in face movement that flat screens and printed photos cannot replicate. Measures inter-frame displacement patterns for 3D presence proof.

Face Micro-Movements

Subtle involuntary facial movements that distinguish live faces from static images. Analyzes frame-to-frame landmark drift for natural micro-motion signatures.

Face Movement Challenge

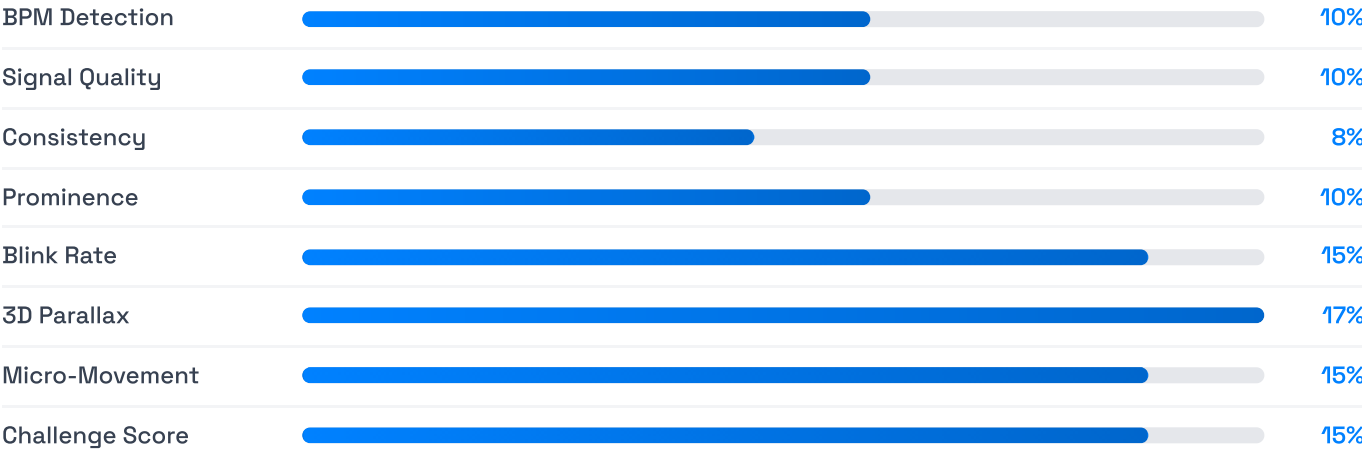
Random prompts to turn head left/right, look up/down, nod, or blink on command. Verifies real-time control and responsiveness of a live subject.

Signal Quality & Consistency

Continuous monitoring of rPPG signal quality, BPM consistency across windows, and frequency peak prominence to ensure reliable measurements throughout the session.

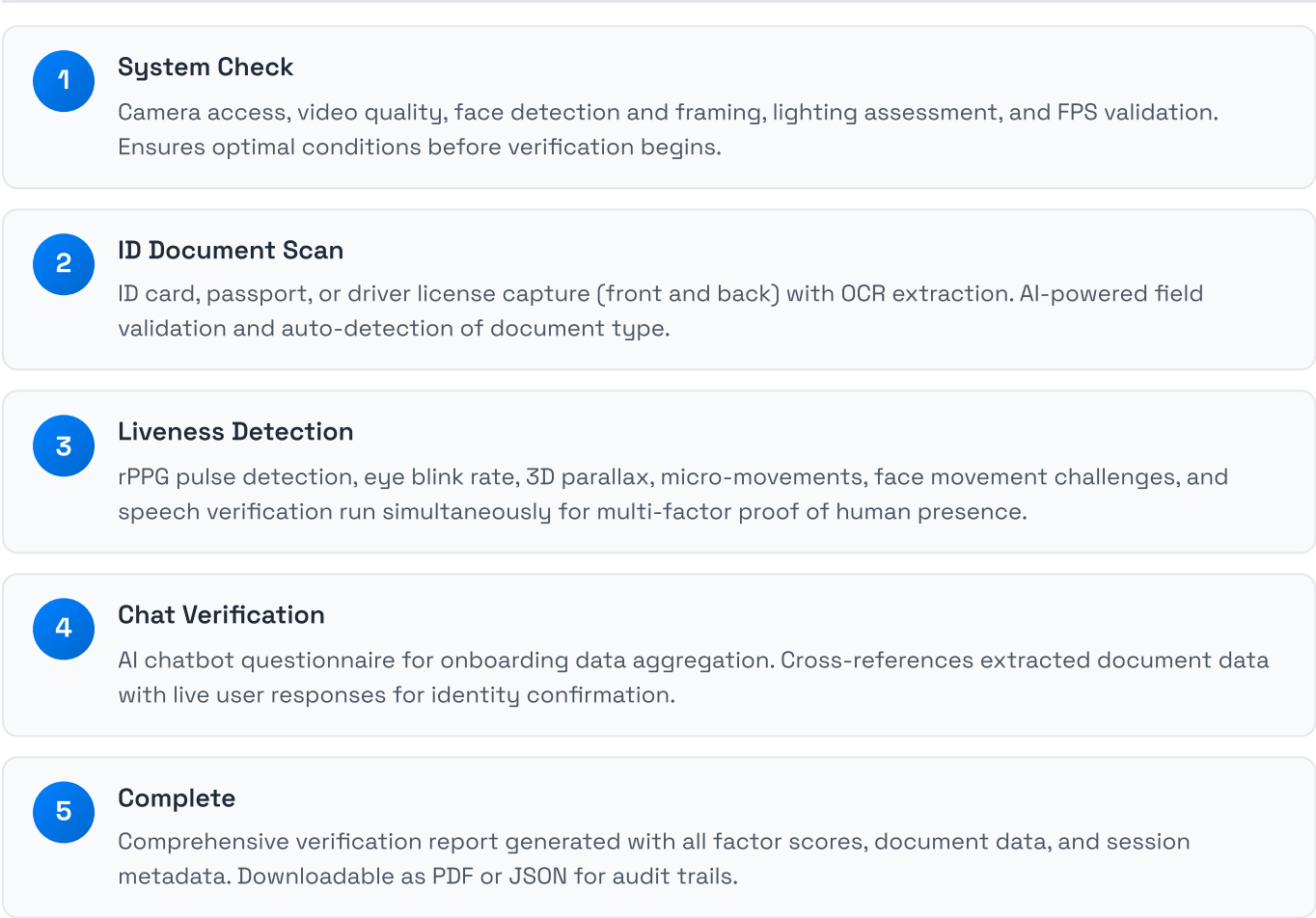
8-Factor Liveness Analysis

Each factor contributes a weighted score to the overall liveness confidence. A minimum threshold of 60% combined confidence is required for verification.



Total: **100%** weighted across 8 independent signals. Behavioral factors (blink, parallax, micro-movement, challenge) account for 62% of the score, making spoofing extremely difficult.

5-Step Verification Workflow



Privacy by Design

BlockSign Verify is built on a zero-storage architecture. No video frames, images, or biometric templates are ever stored or transmitted. All processing happens in real-time with ephemeral data that is discarded immediately after the session ends.

Principle	Implementation
No Video Storage	All frames processed in memory only, never written to disk
No Biometric Templates	No facial embeddings, voiceprints, or fingerprints saved
Ephemeral Processing	Time-series data exists only during active session
Session Isolation	Each verification session is independent and fully discarded
GDPR & CCPA	Full compliance with data protection regulations
Audit Reports	Verification outcome reports without raw biometric data

Technology Stack

Frontend

- React + TypeScript
- face-api.js (68-point landmarks)
- Web Speech API
- Web Audio API (volume)
- MediaDevices API (camera)

Backend

- Node.js + Express
- WebSocket (real-time)
- FFT signal processing
- rPPG analysis pipeline
- Gemini AI (OCR)

Architecture

- WebSocket streaming
- Ephemeral session data
- Sub-second analysis
- No database required
- REST API ready

Enterprise Integration

Reporting & Audit

- PDF verification reports with full breakdown
- JSON export for system integration
- 8-factor confidence scoring
- Session metadata and timestamps

User Experience

- Auto-detection and auto-capture
- Real-time visual feedback overlays
- Volume monitoring during speech
- Guided step-by-step workflow

Integration

- REST API for backend integration
- WebSocket for real-time streaming
- Customizable verification flows
- White-label ready

Configuration

- Customizable question catalogs
- Adjustable confidence thresholds
- Configurable challenge types
- Multi-language support ready

Proof of Human. Built for the AI Era.

Ready to secure your critical processes?

Contact us to learn more about BlockSign Verify.

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